

Q1 Handling ArithmeticException

10 marks

Write a Java program to divide two integers entered by the user. If the denominator is zero, handle the exception using a try-catch block and display an appropriate error message.

Your Code

```
1 import java.util.Scanner;
2 public class Main
3 {
4     public static void main(String[] args)
5     {
6         int a=10;
7         int b=0;
8         try
9         {
10            int result=a/b;
11            System.out.println("Result="+result);
12        }
13        catch(ArithmeticException e)
14        {
15            System.out.println("Error:Division by 0 is not allowed");
16        }
17    }
18 }
```

Input

stdin input

Output

Error:Division by 0 is not allowed

Q2 Handling ArrayIndexOutOfBoundsException

10 marks

Write a Java program to create an integer array of five elements. Ask the user to enter an index and display the corresponding element. If the index is outside the valid range, handle the exception and display an error message.

Your Code

```
1 java.util.Scanner;
2 class ArrayException
3
4 public static void main(String[] args)
5
6 int[] num={10,20,30,40,50};
7 int index=3;
8 try
9 {
10     System.out.println("Element at index "+index+" is "+num[index]);
11 }
12 catch(ArrayIndexOutOfBoundsException e)
13 {
14     System.out.println("Error: The index entered is outside the valid range");
15 }
16 }
```

Input

stdin input

Output

Element at index 3 is 40

Q3 Handling NumberFormatException

Write a Java program to accept a number as a string and convert it into an integer using Integer.parseInt(). If the input is not a valid integer, handle the exception and display an appropriate message.

Your Code

```
1 import java.util.Scanner;
2 public class main
3 {
4     public static void main(String[] args)
5     {
6         String input="123";
7         try
8         {
9             int num=Integer.parseInt(input);
10            System.out.println("Converted integer value:"+num);
11        }
12        catch (NumberFormatException e)
13        {
14            System.out.println("Invalid input! please enter a valid in
15        }
16    }
17 }
```

Input

stdin input

Output

Converted integer value:123

Q4 Handling Multiple Exceptions

Write a Java program that performs the following operations:

Accept two integers from the user.

Divide the first number by the second.

Store the result in an array.

Handle both ArithmeticException and ArrayIndexOutOfBoundsException using multiple catch blocks.

```
1 import java.util.Scanner;
2 public class Main
3 {
4     public static void main(String[] args)
5     {
6         int num1=10;
7         int num2=2;
8         int index=2;
9         try
10        {
11            int arr[]=new int[3];
12            int result=num1/num2;
13            arr[index]=result;
14            System.out.println("Stored result in array successfully");
15            System.out.println("arr[0]="+arr[0]+", arr[1]="+arr[1]+", a
16        }
17        catch (ArithmeticException e)
18        {
19            System.out.println("ArithmeticException:Division by 0 is no
20        }
21        catch (ArrayIndexOutOfBoundsException e)
22        {
23            System.out.println("ArrayIndexOutOfBoundsException:Invalid
24        }
25    }
26 }
```

stdin input

Output

Stored result in array successfully
arr[0]=0, arr[1]=0, arr[2]=5